

LOW STYRENE SBR PATENT LANDSCAPE: EMULSION, SOLUTION, FUNCTIONALIZED, AND PROCESS-CONTROL APPROACHES

1. ABSTRACT

This report synthesizes seven primary patents relevant to styrene-butadiene rubber (SBR) synthesis and production, with emphasis on low-styrene embodiments and process routes that can support reduced styrene incorporation. The landscape spans aqueous emulsion polymerization, organic-solution polymerization, functionalized SBR latex preparation, oil-extension of SBR, modified diene polymer design for tire performance, battery-binder SBR microspheres, and continuous-process reaction blocking for SBR elastomers. Across the set, the target material is not always explicitly labeled as low-styrene SBR; instead, relevance arises from patents that disclose SBR preparation, monomer feed control, or SBR-derived materials where styrene content can be tuned.

The strongest direct synthesis disclosures are the emulsion-polymerized SBR patents (US8530017B2, US9902837B2, US12129321B2, and EP0295519A1) and the solution-polymerized SSBR patent (EP2657262B1). US8530017B2 teaches radical emulsion polymerization of styrene and butadiene using reactive emulsifiers, with styrene feed explicitly in the 10-50 wt% range and butadiene in the 45-85 wt% range. US9902837B2 and US12129321B2 both disclose aqueous emulsion systems with water as the polymerization medium and later coagulation/washing steps; US9902837B2 further shows functional monomer incorporation and oil extension, while US12129321B2 focuses on porous cross-linked SBR microspheres for lithium-ion battery binders. EP2657262B1 is the key non-aqueous route, disclosing anionically initiated solution polymerization in hexane with 70/30 butadiene/styrene monomer feed and subsequent oil extension/steam stripping.

Overall, the patent set indicates that low-styrene SBR development is most plausibly achieved by controlling styrene feed ratio in emulsion or solution polymerization, using reactive emulsifiers or functional monomers to preserve performance at lower styrene levels, and applying downstream oil extension, coagulation, stripping, or crosslinking to tailor processability and end-use properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8530017B2
- Patent Title: Method for preparing styrene-butadiene copolymer using reactive emulsifier and ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses radical emulsion polymerization of styrene and 1,3-butadiene using a reactive emulsifier that also participates in the polymerization. The process is carried out in a pressurized aqueous reactor, followed by termination, stripping, and drying to obtain styrene-butadiene copolymers and related functionalized variants. - Water: 1600 mL — Example 1 - Potassium rosinate: 11 g — Example 1 - Fatty acid potassium salt: 18 g — Example 1 - Potassium chloride: 1.5 g — Example 1 - Styrene: 260 g — Example 1 - Methylcarboxylate methacrylate: 30 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - n-Dodecyl mercaptan: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - Reaction time: 8 hours — Example 1 - Reaction temperature: 10 °C — Example 1 - 5-L pressurized reactor: 5 L — Example 1 - Diethylhydroxyamine: 1.0 g — Example 1 - Weight average molecular weight: 860,000 g/mol — Example 1 - Water: 1600 mL — Example 2 - Potassium rosinate: 11 g — Example 2 - Fatty acid potassium salt: 18 g — Example 2 - Potassium chloride: 1.5 g — Example 2 - Styrene: 280 g — Example 2 - Polyoxyethylene phosphate methacrylate: 10 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - EDTA: 0.5 g — Example 2 - Sodium hydrosulfite: 0.3 g — Example 2 - Ferrous sulfate: 0.1 g — Example 2 - Sodium formaldehyde sulfonate: 0.4 g — Example 2 - n-Dodecyl mercaptan: 1.0 g — Example 2 - Methane hydroperoxide: 0.6 g — Example 2 - Reaction time: 8 hours — Example 2 - Reaction temperature: 10 °C — Example 2 - 5-L pressurized reactor: 5 L — Example 2 - Diethylhydroxyamine: 1.0 g — Example 2 - Weight average molecular weight: 820,000 g/mol — Example 2 - Water: 1600 mL — Example 3 - Potassium rosinate: 11 g — Example 3 - Fatty acid potassium salt: 18 g — Example 3 - Potassium chloride: 1.5 g — Example 3 - Styrene: 260 g — Example 3 - Polyoxyethylene phosphate methacrylate: 30 g — Example 3 - 1,3-Butadiene: 710 g — Example 3 - EDTA: 0.5 g — Example 3 - Sodium hydrosulfite: 0.3 g — Example 3 - Ferrous sulfate: 0.1 g — Example 3 - Sodium formaldehyde sulfonate: 0.4 g — Example 3 - n-Dodecyl mercaptan: 1.5 g — Example 3 - Methane hydroperoxide: 0.6 g — Example 3 - Reaction time: 8 hours — Example 3 - Reaction temperature: 10 °C — Example 3 - 5-L pressurized reactor: 5 L — Example 3 - Diethylhydroxyamine: 1.0 g — Example 3 - Weight average molecular weight: 824,000 g/mol —

Example 3 - Water: 1600 mL — Example 4 - Potassium rosinate: 11 g — Example 4 - Fatty acid potassium salt: 18 g — Example 4 - Potassium chloride: 1.5 g — Example 4 - Styrene: 260 g — Example 4 - Allyl alcohol ethoxylate: 30 g — Example 4 - 1,3-Butadiene: 710 g — Example 4 - EDTA: 0.5 g — Example 4 - Sodium hydrosulfite: 0.3 g — Example 4 - Ferrous sulfate: 0.1 g — Example 4 - Sodium formaldehyde sulfonate: 0.4 g — Example 4 - n-Dodecyl mercaptan: 1.0 g — Example 4 - Methane hydroperoxide: 0.6 g — Example 4 - Reaction time: 8 hours — Example 4 - Reaction temperature: 10 °C — Example 4 - 5-L pressurized reactor: 5 L — Example 4 - Diethylhydroxyamine: 1.0 g — Example 4 - Weight average molecular weight: 850,000 g/mol — Example 4 - Water: 1600 mL — Example 5 - Potassium rosinate: 11 g — Example 5 - Fatty acid potassium salt: 18 g — Example 5 - Potassium chloride: 1.5 g — Example 5 - Styrene: 260 g — Example 5 - Allyl alcohol polyethylene oxide phosphate: 30 g — Example 5 - 1,3-Butadiene: 710 g — Example 5 - EDTA: 0.5 g — Example 5 - Sodium hydrosulfite: 0.3 g — Example 5 - Ferrous sulfate: 0.1 g — Example 5 - Sodium formaldehyde sulfonate: 0.4 g — Example 5 - n-Dodecyl mercaptan: 1.0 g — Example 5 - Methane hydroperoxide: 0.6 g — Example 5 - Reaction time: 8 hours — Example 5 - Reaction temperature: 10 °C — Example 5 - 5-L pressurized reactor: 5 L — Example 5 - Diethylhydroxyamine: 1.0 g — Example 5 - Weight average molecular weight: 900,000 g/mol — Example 5 - Water: 1600 mL — Example 6 - Potassium rosinate: 11 g — Example 6 - Fatty acid potassium salt: 18 g — Example 6 - Potassium chloride: 1.5 g — Example 6 - Styrene: 260 g — Example 6 - Allyl polyoxyethylene glycol methacrylate: 30 g — Example 6 - 1,3-Butadiene: 710 g — Example 6 - EDTA: 0.5 g — Example 6 - Sodium hydrosulfite: 0.3 g — Example 6 - Ferrous sulfate: 0.1 g — Example 6 - Sodium formaldehyde sulfonate: 0.4 g — Example 6 - n-Dodecyl mercaptan: 1.0 g — Example 6 - Methane hydroperoxide: 0.6 g — Example 6 - Reaction time: 8 hours — Example 6 - Reaction temperature: 10 °C — Example 6 - 5-L pressurized reactor: 5 L — Example 6 - Diethylhydroxyamine: 1.0 g — Example 6 - Weight average molecular weight: 950,000 g/mol — Example 6 - Water: 1600 mL — Example 7 - Potassium rosinate: 11 g — Example 7 - Fatty acid potassium salt: 18 g — Example 7 - Potassium chloride: 1.5 g — Example 7 - Styrene: 260 g — Example 7 - Polyoxypropylene glycol phosphate acrylate: 30 g — Example 7 - 1,3-Butadiene: 710 g — Example 7 - EDTA: 0.5 g — Example 7 - Sodium hydrosulfite: 0.3 g — Example 7 - Ferrous sulfate: 0.1 g — Example 7 - Sodium formaldehyde sulfonate: 0.4 g — Example 7 - n-Dodecyl mercaptan: 1.0 g — Example 7 - Methane hydroperoxide: 0.6 g — Example 7 - Reaction time: 8 hours — Example 7 - Reaction temperature: 10 °C — Example 7 - 5-L pressurized reactor: 5 L — Example 7 - Diethylhydroxyamine: 1.0 g — Example 7 - Weight average molecular weight: 845,000 g/mol — Example 7 - Medium & Water Role: Water is part of the emulsion polymerization medium (1600 mL water in Examples 1-7), and later water is also used in coagulation/washing of the latex-derived crumbs before drying. - styrene content below conventional SBR grades: 10-50 wt % based on the weight of the total reactants

Medium & Water Role - Water is part of the emulsion polymerization medium (1600 mL water in Examples 1-7), and later water is also used in coagulation/washing of the latex-derived crumbs before drying. - Core reaction medium: Aqueous emulsion / latex polymerization medium

styrene content below conventional SBR grades - 10-50 wt % based on the weight of the total reactants

Relevant Experimental Evidence - Example 1: Emulsion polymerization with methylcarboxylate methacrylate reactive emulsifier produced SBR copolymer with Mw 860,000 g/mol. - Example 2: Increasing styrene to 280 g with polyoxyethylene phosphate methacrylate gave Mw 820,000 g/mol. - Example 3: Higher reactive emulsifier loading (30 g) with styrene 260 g yielded Mw 824,000 g/mol. - Example 4: Allyl alcohol ethoxylate functionalized copolymer was prepared in the same aqueous emulsion process. - Example 7: Polyoxypropylene glycol phosphate acrylate functionalized copolymer gave Mw 845,000 g/mol.

Technical Relevance This patent is directly relevant because it discloses emulsion SBR synthesis with explicit styrene feed levels and reactive emulsifier systems that can be used to tune styrene incorporation. The disclosed 10-50 wt% styrene range is especially pertinent to low-styrene SBR design.

Patent 2

- Patent Number: EP2657262B1
- Patent Title: Triglyceride containing solution polymerization prepared styrene/butadiene ...
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionically initiated solution polymerization of styrene and 1,3-butadiene in hexane to form SSBR cement, followed by termination and oil extension. The polymer cement is then steam stripped and dried to recover the elastomer composite. - Reactor volume: 2000 liter — Example I - Reactor temperature: 63 to 71 °C — Example I - Internal pressure: 97 to 186 kPa — Example I - Total monomers: 12 weight percent — Example I - 1,3-Butadiene: 70 weight percent of total monomers — Example I - Styrene: 30 weight percent of total monomers — Example I - TMEDA: 0.12 pphm — Example I - SMT: 0.0035 pphm — Example I - n-BuLi: 0.025 pphm — Example I - Conversion: 90 to 95 percent — Example I - Polystay K: 0.5 pphm — Example I - Soybean oil: 36.9 pphr — Example I - Petroleum oil: 36.8 pphr — Example I - Mooney viscosity (23°C): about 107 — Example I - Mooney viscosity (23°C): about 52.8 — Polymer X - Mooney viscosity (23°C): about 40 — Polymer Y - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: 30 weight percent of total monomers

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent solution polymerization in hexane

styrene content below conventional SBR grades - 30 weight percent of total monomers

Relevant Experimental Evidence - Example I: Base SSBR was prepared by anionically initiated solution polymerization in hexane with 70/30 butadiene/styrene feed. - Example I: Polymerization was terminated at 90-95% conversion and the cement had Mooney viscosity about 107. - Example I: Petroleum oil extension reduced Mooney viscosity to about 52.8 after steam stripping. - Example I: Soybean oil extension reduced Mooney viscosity further to about 40.

Technical Relevance This patent is relevant because it discloses a solution SBR route with explicit styrene/butadiene feed ratio control, which is a direct lever for lowering styrene content. It also shows how oil extension and cement handling can be integrated into SSBR manufacture.

Patent 3

- Patent Number: US6313232B1
- Patent Title: Continuous addition of tin halides to solution SBR
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2001
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses continuous hydrocarbon-solution polymerization of butadiene and styrene using an organolithium initiator, with tin compound addition during polymerization and again after withdrawal of the polymer cement. The process is aimed at improving hysteresis and rolling resistance of the resulting diene polymers, including butadiene-styrene copolymers. - Polymer cement sample size: approximately 300-400 g — Comparative Example - Bu_3SnCl solution concentration: 10% by weight in hexane — Comparative Example - SnCl_4 solution concentration: 10% by weight in hexane — Comparative Example - Active lithium equivalence: 0.04 g per 100 g of monomer — Comparative Example - Mixing time: 15-30 minutes — Comparative Example - Natural rubber: 25 parts — Compounding - Carbon black: 48.5 parts — Compounding - ZnO : 25 parts — Compounding - Wax: 1.0 parts — Compounding - Hydrocarbon resin: 2.0 parts — Compounding - Amine antioxidant: 0.95 parts — Compounding - Stearic acid: 2.0 parts — Compounding - Aro oil: 6 parts — Compounding - Nap oil: 5 parts — Compounding - CBS accelerator: 1.4 parts — Compounding - DPG accelerator: 0.15 parts — Compounding - Sulfur: 1.5 parts — Compounding - Brabender mixing time: 6 minutes — Compounding - Brabender mixing temperature: 90 °C — Compounding - Second mixing time: 1 minute 30 seconds — Compounding - Second mixing temperature: 105 °C — Compounding - Dynastat test temperature: 50 °C — Evaluation - Rheometrics RDA 700 temperature: 25 °C — Evaluation - Strain increment: 0.25% — Evaluation - Maximum strain: 14.7% — Evaluation - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. -
Core reaction medium: Hydrocarbon solvent solution polymerization

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Example: Polymer cement was treated with Bu₃SnCl or SnCl₄ in hexane and then drum dried. - Comparative Example: Dynastat results showed improved hysteresis loss for tin-treated samples versus control. - Comparative Example: Strain sweep showed reduced Payne effect for tin-treated samples. - Comparative Example: Repeating the process with different tin compounds gave substantially similar results.

Technical Relevance This patent is relevant as a solution-polymerization process for butadiene-styrene copolymers with explicit continuous-process control. Although it does not target low styrene directly, it provides a manufacturing route where styrene feed and polymer architecture can be tuned.

Patent 4

- Patent Number: US9902837B2
- Patent Title: Oil extended functionalized styrene-butadiene copolymer
- Assignee: Arlanxeo Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of standard and functionalized SBR latex in water, followed by blending with extension oil, coagulation, washing, and drying. The functionalized route introduces glycidyl methacrylate at the beginning of polymerization to improve silica compatibility and tire performance. - Water: 8,700 g — Example 1 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 1 - 1,3-Butadiene: 3,830 g — Example 1 - Styrene: 1,486 g — Example 1 - Pinane hydroperoxide: 2.5 g — Example 1 - Activator batch: 390 g — Example 1 - t-Dodecyl mercaptan: 10 g — Example 1 - Reaction temperature: 7 °C — Example 1 - Conversion: 60% — Example 1 - Polymerization duration: about 7 hours — Example 1 - Extension oil: 37.5 phr — Example 1 - Coagulation temperature: 65 °C — Example 1 - Drying time: 18 hours — Example 1 - Water: 8,500 g — Example 4 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 4 - Buffering system pH: 8.7 — Example 4 - 1,3-Butadiene: 3,830 g — Example 4 - Styrene: 1,242 g — Example 4 - GMA/styrene blend: 106/133 g — Example 4 - Pinane hydroperoxide: 2.5 g — Example 4 - Activator batch: 390 g — Example 4 - t-Dodecyl mercaptan: 10 g — Example 4 - Reaction temperature: 7 °C — Example 4 - Conversion: 60% — Example 4 - Polymerization duration: about 5 hours — Example 4 - Final latex pH: 9.2 — Example 4 - Extension oil: 37.5 phr — Example 4 - Coagulation temperature: 65 °C — Example 4 - Drying

time: 18 hours — Example 4 - Bound styrene content: 21.0% wt — Example 6 - cis-1,4 content: 9.5% wt — Example 6 - trans-1,4 content: 76.3% wt — Example 6 - vinyl 1,2 content: 14.2% wt — Example 6 - Mooney viscosity (ML 1+4, 100 °C): 45.8 — Example 6 - Average molecular weight (Mw): 356,000 g/mol — Example 6 - Rubber: 96.2 phr — Table 2/3 - Buna CB 24 (polybutadiene): 30.0 phr — Table 2/3 - Carbon black (N-339): 75.0 phr — Table 2 - Oil: 11.3 phr — Table 2/3 - Silica: 70.0 phr — Table 3 - Silane: 5.6 phr — Table 3 - Sulfur: 2.0 phr — Table 2 - TBBS: 1.3 phr — Table 2 - DPG: 0.3 phr — Table 2 - Sulfur: 1.6 phr — Table 3 - TBBS: 1.0 phr — Table 3 - DPG: 2.0 phr — Table 3 - Medium & Water Role: Water is the polymerization medium in the emulsion latex process, and later water is used for coagulation/washing of the crumb before drying. - styrene content below conventional SBR grades: 21.0% wt

Medium & Water Role - Water is the polymerization medium in the emulsion latex process, and later water is used for coagulation/washing of the crumb before drying. - Core reaction medium: Aqueous emulsion / latex polymerization medium

styrene content below conventional SBR grades - 21.0% wt

Relevant Experimental Evidence - Example 1: Standard SBR latex was prepared by aqueous emulsion polymerization and oil-extended with 37.5 phr extension oil. - Example 4: Functionalized SBR latex was prepared by adding GMA/styrene blend at the start of emulsion polymerization. - Example 6: Epoxidized soybean oil extension gave a functionalized SBR with 21.0 wt% bound styrene and Mw 356,000 g/mol. - Example 6: Carbon black and silica compound data showed improved abrasion and tan delta behavior relative to controls.

Technical Relevance This patent is highly relevant because it discloses aqueous emulsion SBR synthesis with explicit styrene feed amounts and a functionalized variant that can be used to tune performance at lower styrene levels. The 21.0 wt% bound styrene example is particularly pertinent to low-styrene SBR embodiments.

Patent 5

- Patent Number: US12129321B2
- Patent Title: Cross-linked SBR microsphere binder and preparation method thereof and lithium-...
- Assignee: Zhuhai Cosmx Battery Co., Ltd.
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of porous cross-linked SBR microsphere binders by emulsion-type polymerization in deionized water with pore-forming agents, emulsifiers, oxidizing/initiator systems, and post-polymerization

demulsification, extraction, and washing. The process is tailored for lithium-ion battery binders rather than conventional tire SBR. - Sodium hydroxide aqueous solution: 8% mass fraction — Example 1 - Pretreatment temperature: 20 °C — Example 1 - Pretreatment time: 4 h — Example 1 - Deionized water: 200 parts — Example 1 - Styrene: 25 parts — Example 1 - n-Heptane: 80 parts — Example 1 - n-Propanol: 20 parts — Example 1 - Methanol: 20 parts — Example 1 - Glycerin: 40 parts — Example 1 - Xylene: 40 parts — Example 1 - Butadiene: 75 parts — Example 1 - Dodecyl mercaptan: 0.16 parts — Example 1 - Disproportionated rosin acid soap: 4.62 parts — Example 1 - Sodium methylene bisnaphthalene sulfonate: 0.15 parts — Example 1 - Sodium dithionite dihydrate: 0.025 parts — Example 1 - Potassium phosphate: 0.24 parts — Example 1 - Ferrous sulfate: 0.01 parts — Example 1 - EDTA-disodium salt: 0.01 parts — Example 1 - Cumene hydrogen peroxide: 0.06 parts — Example 1 - Reactor pressure: 0.1 MPa — Example 1 - Polymerization temperature: 8 °C — Example 1 - Stirring speed: 100 r/min — Example 1 - Polymerization time: 7 h — Example 1 - Sodium dimethyl dithiocarbamate: 0.1 parts — Example 1 - Sodium nitrite: 0.02 parts — Example 1 - Sodium polysulfide: 0.02 parts — Example 1 - Polyvinylamine: 0.02 parts — Example 1 - 20% sodium chloride solution: used for demulsification — Example 1 - 0.5% dilute sulfuric acid: used for demulsification — Example 1 - Washing temperature: 50 °C — Example 1 - Diethyl ether: extraction solvent at 50 °C — Example 1 - Particle size: 10 nm to 100 nm — Example 1 - Porosity: 0.01% to 3% — Example 1 - Pore diameter: greater than 0 nm and less than or equal to 10 nm — Example 1 - Sodium hydroxide aqueous solution: 10% mass fraction — Example 2 - Pretreatment temperature: 40 °C — Example 2 - Pretreatment time: 10 min — Example 2 - Deionized water: 400 parts — Example 2 - Styrene: 50 parts — Example 2 - n-Octane: 100 parts — Example 2 - Cyclohexane: 50 parts — Example 2 - Cyclohexanol: 30 parts — Example 2 - 2-Ethylbutanol: 30 parts — Example 2 - n-Butyl ether: 30 parts — Example 2 - Benzene: 50 parts — Example 2 - Acetonitrile: 10 parts — Example 2 - Butadiene: 150 parts — Example 2 - Polyethylene glycol methyl methacrylate: 10 parts — Example 2 - Polyethylene glycol diacrylate: 2 parts — Example 2 - Methyl methacrylate: 8 parts — Example 2 - Dithiodiisopropyl xanthate: 0.352 parts — Example 2 - Fatty acid soap: 10.16 parts — Example 2 - Naphthalenesulfonic acid: 0.33 parts — Example 2 - Sodium dithionite dihydrate: 0.082 parts — Example 2 - Potassium phosphate: 0.908 parts — Example 2 - Ferrous sulfate: 0.022 parts — Example 2 - EDTA-disodium salt: 0.456 parts — Example 2 - Hydrogen peroxide p-menthane: 0.24 parts — Example 2 - Sodium dimethyl dithiocarbamate: 0.22 parts — Example 2 - Sodium nitrite: 0.075 parts — Example 2 - Sodium polysulfide: 0.075 parts — Example 2 - Polyvinylamine: 0.075 parts — Example 2 - Sodium hydroxide aqueous solution: 12% mass fraction — Example 3 - Pretreatment temperature: 30 °C — Example 3 - Pretreatment time: 2 h — Example 3 - Deionized water: 600 parts — Example 3 - Styrene: 75 parts — Example 3 - Ethyl acetate: 100 parts — Example 3 - Acetone: 50 parts — Example 3 - Diethyl phthalate: 50 parts — Example 3 - Cyclohexane: 100 parts — Example 3 - Isooctyl alcohol: 20 parts — Example 3 - n-Butyl ether: 30 parts — Example 3 - Succinonitrile: 20 parts — Example 3 - Chloroform: 20 parts — Example 3 - Urea: 5 parts — Example 3 - Polystyrene: 3 parts — Example 3 - Polyethylene glycol: 2 parts — Example 3 - Butadiene: 225 parts — Example 3 - Polyethylene glycol monoallyl ether: 6 parts — Example 3 - Polyethylene glycol methyl acrylate: 8 parts — Example 3 - Ethyl methacrylate: 8 parts — Example 3 - Acrylic acid: 10 parts — Example 3 - Dodecyl mercaptan: 0.528 parts — Example 3 - Disproportionated rosin acid soap: 15.25 parts — Example 3 - Sodium methylene bisnaphthalene sulfonate: 0.495 parts — Example 3 - Sodium dithionite dihydrate:

0.123 parts — Example 3 - Potassium phosphate: 1.362 parts — Example 3 - Ferrous sulfate: 0.033 parts — Example 3 - EDTA-disodium salt: 0.684 parts — Example 3 - Cumene hydrogen peroxide: 0.36 parts — Example 3 - Reactor pressure: 0.25 MPa — Example 3 - Polymerization temperature: 7 °C — Example 3 - Stirring speed: 110 r/min — Example 3 - Polymerization time: 9 h — Example 3 - Sodium dimethyl dithiocarbamate: 0.33 parts — Example 3 - Sodium polysulfide: 0.13 parts — Example 3 - Polyvinylamine: 0.10 parts — Example 3 - Sodium nitrite: 0.10 parts — Example 3 - 24% sodium chloride solution: used for demulsification — Example 3 - 0.5% dilute sulfuric acid: used for demulsification — Example 3 - Washing temperature: 60 °C — Example 3 - Methanol: extraction solvent at 65 °C — Example 3 - Particle size: 60 nm to 200 nm — Example 3 - Porosity: 8% to 12% — Example 3 - Pore diameter: greater than 0 nm and less than or equal to 60 nm — Example 3 - Sodium hydroxide aqueous solution: 14% mass fraction — Example 4 - Pretreatment temperature: 25 °C — Example 4 - Pretreatment time: 1 h — Example 4 - Deionized water: 800 parts — Example 4 - Styrene: 100 parts — Example 4 - Fatty acid methyl ester: 150 parts — Example 4 - Toluene: 345 parts — Example 4 - Isopropanol: 50 parts — Example 4 - Di-n-amyl ether: 50 parts — Example 4 - Methyl ethyl ketone: 50 parts — Example 4 - Acetonitrile: 50 parts — Example 4 - Chloroform: 30 parts — Example 4 - Urea: 20 parts — Example 4 - Polyethylene glycol: 55 parts — Example 4 - Butadiene: 300 parts — Example 4 - Acrylonitrile: 10 parts — Example 4 - Methyl acrylate: 10 parts — Example 4 - Polyethylene glycol monomethyl ether methyl methacrylate: 20 parts — Example 4 - 2-Ethylhexyl acrylate: 10 parts — Example 4 - Dodecyl mercaptan: 0.72 parts — Example 4 - Fatty acid soap: 20.79 parts — Example 4 - Sodium methylenediphenyl sulfonate: 0.675 parts — Example 4 - Sodium dithionite dihydrate: 0.18 parts — Example 4 - Potassium phosphate: 2.03 parts — Example 4 - Ferrous sulfate: 0.045 parts — Example 4 - EDTA-disodium salt: 1.13 parts — Example 4 - Cumene hydrogen peroxide: 0.54 parts — Example 4 - Reactor pressure: 0.3 MPa — Example 4 - Polymerization temperature: 5 °C — Example 4 - Stirring speed: 130 r/min — Example 4 - Polymerization time: 12 h — Example 4 - Sodium dimethyl dithiocarbamate: 0.45 parts — Example 4 - Sodium nitrite: 0.2 parts — Example 4 - Sodium nitrite sodium polysulfide: 0.15 parts — Example 4 - Sodium nitrite polyvinylamine: 0.15 parts — Example 4 - 30% sodium chloride solution: used for demulsification — Example 4 - 0.5% dilute sulfuric acid: used for demulsification — Example 4 - Washing temperature: 55 °C — Example 4 - Acetone and benzene: extraction solvents at 70 °C — Example 4 - Particle size: 80 nm to 300 nm — Example 4 - Porosity: 12% to 20% — Example 4 - Pore diameter: greater than 0 nm and less than or equal to 80 nm — Example 4 - Sodium hydroxide aqueous solution: 16% mass fraction — Example 5 - Pretreatment temperature: 35 °C — Example 5 - Pretreatment time: 3 h — Example 5 - Deionized water: 500 parts — Example 5 - Styrene: 62.5 parts — Example 5 - Ethyl acetate: 100 parts — Example 5 - Acetone: 100 parts — Example 5 - Diethyl phthalate: 100 parts — Example 5 - n-Heptane: 50 parts — Example 5 - Dodecanol: 50 parts — Example 5 - Succinonitrile: 15 parts — Example 5 - 1,2-Dichloroethane: 10 parts — Example 5 - Amino acid: 10 parts — Example 5 - Polystyrene: 5 parts — Example 5 - Polyacrylic acid: 5 parts — Example 5 - Polyvinyl alcohol: 5 parts — Example 5 - Butadiene: 187.5 parts — Example 5 - Polyethylene glycol monomethyl ether methyl acrylate: 8 parts — Example 5 - Polyethylene glycol monoallyl ether: 8 parts — Example 5 - Acrylonitrile: 9 parts — Example 5 - Dodecyl mercaptan: 0.44 parts — Example 5 - Disproportionated rosin acid soap: 12.71 parts — Example 5 - Sodium methylene bisnaphthalene sulfonate: 0.4125 parts — Example 5 - Sodium dithionite dihydrate: 0.1025 parts — Example 5 - Potassium phosphate:

1.135 parts — Example 5 - Ferrous sulfate: 0.0275 parts — Example 5 - EDTA-disodium salt: 0.57 parts — Example 5 - Hydrogen peroxide p-menthane: 0.30 parts — Example 5 - Reactor pressure: 0.15 MPa — Example 5 - Polymerization temperature: 7 °C — Example 5 - Stirring speed: 120 r/min — Example 5 - Polymerization time: 8 h — Example 5 - Sodium dimethyl dithiocarbamate: 0.275 parts — Example 5 - Sodium nitrite: 0.1 parts — Example 5 - Sodium polysulfide: 0.08 parts — Example 5 - Polyvinylamine: 0.1 parts — Example 5 - 26% sodium chloride solution: used for demulsification — Example 5 - 0.5% dilute sulfuric acid: used for demulsification — Example 5 - Washing temperature: 70 °C — Example 5 - Acetone: extraction solvent at 55 °C — Example 5 - Particle size: 90 nm to 400 nm — Example 5 - Porosity: 29% to 35% — Example 5 - Pore diameter: greater than 0 nm and less than or equal to 90 nm — Example 5 - Sodium hydroxide aqueous solution: 9% mass fraction — Example 6 - Pretreatment temperature: 20 °C — Example 6 - Pretreatment time: 2.5 h — Example 6 - Deionized water: 300 parts — Example 6 - Styrene: 37.5 parts — Example 6 - Fatty acid methyl ester: 100 parts — Example 6 - Diethyl phthalate: 200 parts — Example 6 - n-Octane: 100 parts — Example 6 - Di-n-amyl ether: 100 parts — Example 6 - Methyl ethyl ketone: 50 parts — Example 6 - Acetonitrile: 50 parts — Example 6 - 1,2-Dichloroethane: 70 parts — Example 6 - Amino acid: 10 parts — Example 6 - Polystyrene: 5 parts — Example 6 - Polyethylene glycol: 10 parts — Example 6 - Polyacrylonitrile: 5 parts — Example 6 - Butadiene: 112.5 parts — Example 6 - Polyethylene glycol methyl methacrylate: 10 parts — Example 6 - Methyl methacrylate: 2 parts — Example 6 - Acrylic acid: 3 parts — Example 6 - Dithiodiisopropyl xanthate: 0.264 parts — Example 6 - Disproportionated rosin acid soap: 7.63 parts — Example 6 - Sodium methylene bisnaphthalene sulfonate: 0.25 parts — Example 6 - Sodium dithionite dehydrate: 0.0615 parts — Example 6 - Potassium phosphate: 0.682 parts — Example 6 - Ferrous sulfate: 0.0165 parts — Example 6 - EDTA-disodium salt: 0.342 parts — Example 6 - Cumene hydrogen peroxide: 0.18 parts — Example 6 - Reactor pressure: 0.25 MPa — Example 6 - Polymerization temperature: 6 °C — Example 6 - Stirring speed: 130 r/min — Example 6 - Polymerization time: 10 h — Example 6 - Sodium dimethyl dithiocarbamate: 0.165 parts — Example 6 - Sodium nitrite: 0.056 parts — Example 6 - Sodium polysulfide: 0.056 parts — Example 6 - Polyvinylamine: 0.056 parts — Example 6 - 25% sodium chloride solution: used for demulsification — Example 6 - 0.5% dilute sulfuric acid: used for demulsification — Example 6 - Washing temperature: 65 °C — Example 6 - Ethylene oxide and methanol: extraction solvents at 60 °C — Example 6 - Particle size: 90 nm to 400 nm — Example 6 - Porosity: 29% to 35% — Example 6 - Pore diameter: greater than 0 nm and less than or equal to 90 nm — Example 6 - Sodium hydroxide aqueous solution: 8% mass fraction — Example 7 - Pretreatment temperature: 28 °C — Example 7 - Pretreatment time: 3.5 h — Example 7 - Deionized water: 400 parts — Example 7 - Styrene: 50 parts — Example 7 - Fatty acid methyl ester: 100 parts — Example 7 - Acetone: 50 parts — Example 7 - Toluene: 200 parts — Example 7 - Isoamyl alcohol: 100 parts — Example 7 - n-Butyl ether: 50 parts — Example 7 - Acetonitrile: 50 parts — Example 7 - 1,2-Dichloroethane: 30 parts — Example 7 - Amino acid: 10 parts — Example 7 - Polyethylene glycol monomethyl ether: 5 parts — Example 7 - Polyvinyl alcohol: 5 parts — Example 7 - Butadiene: 150 parts — Example 7 - Polyethylene glycol methyl methacrylate: 10 parts — Example 7 - Acrylic acid: 4 parts — Example 7 - Methyl methacrylate: 6 parts — Example 7 - Dodecyl mercaptan: 0.352 parts — Example 7 - Disproportionated rosin acid soap: 10.16 parts — Example 7 - Sodium methylene bisnaphthalene sulfonate: 0.33 parts — Example 7 - Sodium dithionite dihydrate: 0.082 parts — Example 7 - Potassium phosphate: 0.908 parts — Example 7 -

Ferrous sulfate: 0.022 parts — Example 7 - EDTA-disodium salt: 0.456 parts — Example 7 - Hydrogen peroxide p-menthane: 0.24 parts — Example 7 - Reactor pressure: 0.25 MPa — Example 7 - Polymerization temperature: 8 °C — Example 7 - Stirring speed: 110 r/min — Example 7 - Polymerization time: 9 h — Example 7 - Sodium dimethyl dithiocarbamate: 0.22 parts — Example 7 - Sodium nitrite: 0.075 parts — Example 7 - Sodium polysulfide: 0.075 parts — Example 7 - Polyvinylamine: 0.075 parts — Example 7 - 28% sodium chloride solution: used for demulsification — Example 7 - 0.5% dilute sulfuric acid: used for demulsification — Example 7 - Washing temperature: 70 °C — Example 7 - Ethanol: extraction solvent at 65 °C — Example 7 - Particle size: 100 nm to 1 µm — Example 7 - Porosity: 20% to 29% — Example 7 - Pore diameter: greater than 0 nm and less than or equal to 100 nm — Example 7 - Medium & Water Role: Water is the polymerization medium in the SBR microsphere preparation, and later water is used for neutralization, demulsification, washing, and removal of salts/solvents during workup. - styrene content below conventional SBR grades: 25 parts styrene in 200 parts deionized water system

Medium & Water Role - Water is the polymerization medium in the SBR microsphere preparation, and later water is used for neutralization, demulsification, washing, and removal of salts/solvents during workup. - Core reaction medium: Aqueous emulsion / latex polymerization medium with pore-forming organic additives

styrene content below conventional SBR grades - 25 parts styrene in 200 parts deionized water system

Relevant Experimental Evidence - Example 1: A porous cross-linked SBR microsphere binder with 10-100 nm particles and 0.01-3% porosity was prepared in water using n-heptane/n-propanol/methanol/glycerin/xylene pore-forming system. - Example 3: A higher-porosity binder (8-12%) with 60-200 nm particles was obtained using a larger water phase and mixed organic extraction. - Example 5: A binder with 29-35% porosity and 90-400 nm particles was produced using hydrogen peroxide p-menthane and acetone extraction. - Example 6: A similar porous binder was prepared using ethylene oxide/methanol extraction and showed 90-400 nm particles with 29-35% porosity. - Example 7: Ethanol extraction yielded a 100 nm to 1 µm porous binder with 20-29% porosity.

Technical Relevance This patent is relevant because it discloses SBR-based microsphere binders made by aqueous polymerization and post-treatment, demonstrating another SBR manufacturing pathway where styrene feed is explicitly controlled. Although the application is battery-focused rather than tire-focused, it provides low-styrene-compatible SBR synthesis and processing conditions.

Patent 6

- Patent Number: US8614272B2
- Patent Title: Tire rubber composition and pneumatic tire
- Assignee: Sumitomo Rubber Industries, Ltd.
- Jurisdiction: US

- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of modified diene polymers by anionic solution polymerization in cyclohexane using a divinylbenzene/n-butyl lithium initiator system, followed by end modification with a tetraglycidyl amine compound and precipitation/drying. The resulting modified diene polymers are then compounded with silica and other tire ingredients for performance evaluation. - 10 ml of 1,3-divinylbenzene hexane solution: 1.6 M — Production Example 1 - 20 ml of n-butyl lithium hexane solution: 1.6 M — Production Example 1 - 0 °C — Production Example 1 - 1 hour — Production Example 1 - 600 ml cyclohexane — Production Example 2 - 0.12 mol styrene — Production Example 2 - 0.8 mol 1,3-butadiene — Production Example 2 - 0.7 mmol N,N,N',N'-Tetramethylethylenediamine — Production Example 2 - 1.5 ml polymerization initiator solution — Production Example 2 - 40 °C — Production Example 2 - 3 hours — Production Example 2 - 1.0 mmol tetraglycidyl-1,3-bisaminomethylcyclohexane — Production Example 2 - 1 hour — Production Example 2 - 3 ml isopropanol — Production Example 2 - 1 g 2,6-tert-butyl-p-cresol — Production Example 2 - 600 ml cyclohexane — Production Example 3 - 0.8 mol 1,3-butadiene — Production Example 3 - 0.7 mmol N,N,N',N'-Tetramethylethylenediamine — Production Example 3 - 1.5 ml polymerization initiator solution — Production Example 3 - 40 °C — Production Example 3 - 3 hours — Production Example 3 - 1.0 mmol tetraglycidyl-1,3-bisaminomethylcyclohexane — Production Example 3 - 1 hour — Production Example 3 - 3 ml isopropanol — Production Example 3 - 1 g 2,6-tert-butyl-p-cresol — Production Example 3 - Mooney viscosity (ML1+4/100 °C): 60 — Production Example 2 - Mooney viscosity (ML1+4/100 °C): 60 — Production Example 3 - Vinyl content: 57 mol % — Production Example 2 - Vinyl content: 20 mol % — Production Example 3 - Banbury mixing temperature: 165 °C — Examples 1 to 6 - Banbury mixing time: 4 minutes — Examples 1 to 6 - Open roll mill temperature: 80 °C — Examples 1 to 6 - Open roll mill time: 4 minutes — Examples 1 to 6 - Vulcanization temperature: 150 °C — Examples 1 to 6 - Vulcanization time: 35 minutes — Examples 1 to 6 - Vulcanization pressure: 25 kgf — Examples 1 to 6 - Rolling resistance test pressure: 230 kPa — Examples 1 to 6 - Rolling resistance load: 3.43 kN — Examples 1 to 6 - Rolling resistance speed: 80 km/h — Examples 1 to 6 - Tread vulcanization temperature: 170 °C — Examples 7 to 24 - Tread vulcanization time: 12 minutes — Examples 7 to 24 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: 0.12 mol styrene — Production Example 2

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent solution polymerization in cyclohexane

styrene content below conventional SBR grades - 0.12 mol styrene — Production Example 2

Relevant Experimental Evidence - Production Example 2: A modified diene polymer was prepared in cyclohexane from styrene and butadiene using a divinylbenzene/n-butyl lithium

initiator and tetraglycidyl modifier. - Production Example 3: A butadiene-only analogue was prepared under the same solution-polymerization conditions without styrene. - Production Example 2: The modified diene polymer had Mooney viscosity 60 and vinyl content 57 mol%. - Production Example 3: The butadiene-only polymer had Mooney viscosity 60 and vinyl content 20 mol%.

Technical Relevance This patent is relevant because it discloses solution polymerization of styrene-butadiene polymers and shows how monomer choice affects vinyl content and downstream tire performance. It is useful as a process reference for controlled styrene incorporation, although it is not a low-styrene SBR patent per se.

Patent 7

- Patent Number: EP0295519A1
- Patent Title: Apparatus for blocking the copolymerization reaction in systems for the ...
- Assignee: Maria Luisa Piani
- Jurisdiction: EP
- Publication Year: 1988
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses an apparatus and control method for blocking copolymerization in continuous-cycle SBR production lines. It does not teach a new polymerization chemistry, but rather a temperature-controlled, remotely adjustable blocking-agent feed arrangement for stopping SBR copolymerization at the desired solid content. - Required temperature: between 5 and 10 °C — Description - Required solid copolymer value: between 24.8% and 25.2% — Description - Analyzed sample value: 24.5% — Description - Analyzed sample value: 25.5% — Description - Blocking-agent flow ratio: 30% and 70% — Description - Medium & Water Role: Water is explicitly part of the continuous SBR feed mixture and polymerization system; the patent is focused on blocking-agent control rather than downstream water workup. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - Water is explicitly part of the continuous SBR feed mixture and polymerization system; the patent is focused on blocking-agent control rather than downstream water workup. - Core reaction medium: Aqueous continuous-cycle SBR polymerization system

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - No worked examples were present; the disclosure is apparatus-focused. - Description: Continuous-cycle SBR production uses butadiene, styrene, soap, and water fed to refrigerated reactors. - Description: Blocking agent is introduced into pipe reactors or reactor inlets/outlets to stop copolymerization at the desired solid content. -

Description: The apparatus allows remote and automated adjustment of blocking-agent feed based on solid content analysis.

Technical Relevance This patent is relevant as a process-control disclosure for continuous SBR manufacture, especially where precise stopping of polymerization is needed to control final composition and solids content. It does not directly disclose low-styrene synthesis, but it is important for industrial SBR production control.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content below conventional SBR grades	Key Findings
US8530017B2	Korea Kumho Petrochemical Co., Ltd.	2013	Water is part of the emulsion polymerization medium (1600 mL water in Examples 1-7), and later water is also used in coagulation/washing of the latex-derived crumbs before drying.	10-50 wt % based on the weight of the total reactants	Example 1: Emulsion polymerization with methylcarboxylate methacrylate reactive emulsifier produced SBR copolymer with M

EP2657262B1	The Goodyear Tire & Rubber Company	2014	No water-related process step disclosed in the extracted evidence.	30 weight percent of total monomers	Example I: Base SSBR was prepared by anionically initiated solution polymerization in hexane with 70/30 butadiene/styren
US6313232B1	Bridgestone Corporation	2001	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Comparative Example: Polymer cement was treated with Bu ₃ SnCl or SnCl ₄ in hexane and then drum dried.

US9902837B2	Arlanxeo Deutschland GmbH	2018	Water is the polymerization medium in the emulsion latex process, and later water is used for coagulation/washing of the crumb before drying.	21.0% wt	Example 1: Standard SBR latex was prepared by aqueous emulsion polymerization and oil-extended with 37.5 phr extension o
US12129321B2	Zhuhai Cosmx Battery Co., Ltd.	2024	Water is the polymerization medium in the SBR microsphere preparation, and later water is used for neutralization, demulsification, washing, and removal of salts/solvents during workup.	25 parts styrene in 200 parts deionized water system	Example 1: A porous cross-linked SBR microsphere binder with 10-100 nm particles and 0.01-3% porosity was prepared in wa

US8614272B2	Sumitomo Rubber Industries, Ltd.	2013	No water-related process step disclosed in the extracted evidence.	0.12 mol styrene — Production Example 2	Production Example 2: A modified diene polymer was prepared in cyclohexane from styrene and butadiene using a divinylben
EP0295519A1	Maria Luisa Piani	1988	Water is explicitly part of the continuous SBR feed mixture and polymerization system; the patent is focused on blocking-agent control rather than downstream water workup.	Not disclosed in extracted evidence	No worked examples were present; the disclosure is apparatus-focused.

SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US9394647B2
- Patent Title: Adhesive composition for organic fiber cords, rubber-reinforcing material using ...
- Assignee: Bridgestone Corporation
- Jurisdiction: US

- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The packet shows a staged polymerization process that includes a styrene-butadiene polymer with styrene and butadiene content ranges, but the evidence is insufficient to establish the patent as primarily about SBR rather than an adhesive cord treatment process.

Supporting Patent 2

- Patent Number: EP3473654B1
- Patent Title: Method for preparing modified conjugated diene-based polymer, and modified ...
- Assignee: LG Chem, Ltd.
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The invention is a process for preparing modified conjugated diene-based polymers broadly, but the packet does not specifically establish SBR as the central target material for this run.

Supporting Patent 3

- Patent Number: US8314185B2
- Patent Title: Vulcanizable rubber mixture and rubber products comprising the same
- Assignee: Continental Reifen Deutschland GmbH
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is centered on functionalized block copolymers with conjugated diene/aromatic vinyl main chains, which are identity-excluded adjacent architectures rather than SBR itself.

Supporting Patent 4

- Patent Number: US20240368325A1
- Patent Title: High Strength Hydrogenated Polymers, And Rubber Compositions Incorporating Same
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about hydrogenated functionalized copolymers and silane-functionalized styrene-butadiene copolymer, which is adjacent to SBR but not the requested low-styrene SBR base material itself.

Supporting Patent 5

- Patent Number: US5395902A

- Patent Title: Dispersion copolymerization in liquid aliphatic hydrocarbons
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about dispersion copolymerization of styrene and conjugated diene in liquid hydrocarbon medium to make random copolymers, but the evidence does not establish that the central invention is SBR rather than a broader random copolymer process.

Supporting Patent 6

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Nippon Zeon Co., Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a diene rubber made from modified and branched styrene-butadiene rubbers, but the central invention is the combined diene rubber structure rather than low-styrene SBR synthesis itself.

Supporting Patent 7

- Patent Number: EP3138859B1
- Patent Title: Method for producing multi-component copolymer
- Assignee: Bridgestone Corporation
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: Although styrene and butadiene are present, the invention is a multicomponent copolymer with ethylene/non-conjugated olefin architecture, not SBR as the central material.

Supporting Patent 8

- Patent Number: US11318223B2
- Patent Title: Antimicrobial adhesives having improved properties
- Assignee: Avery Dennison Corporation
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about antimicrobial adhesives and only mentions low-styrene SBR as a possible solid rubber ingredient, not as the central invention.

Supporting Patent 9

- Patent Number: US11590257B2
- Patent Title: Antimicrobial hydrocolloid dressing containing sequestered peroxide and ...

- Assignee: Sarasota Medical Products, Inc.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This is a hydrocolloid dressing patent that merely mentions low-styrene SBR as a possible thermoplastic elastomer component, not as the invention focus.

Supporting Patent 10

- Patent Number: US20190375909A1
- Patent Title: Foamed adhesive, more particularly pressure-sensitive adhesive, process for the ...
- Assignee: Tesa Se
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about foamed pressure-sensitive adhesives, where randomly copolymerized SBR is only one possible adhesive-base ingredient and not the invention's central material.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US8530017B2 and US9902837B2 both use aqueous emulsion polymerization, but US8530017B2 explicitly claims styrene feed at 10-50 wt% of total reactants, while US9902837B2 Example 6 reports 21.0 wt% bound styrene in a functionalized SBR.
- EP2657262B1 and US6313232B1 are both solution-polymerization disclosures in hydrocarbon solvent, but EP2657262B1 gives a clear 70/30 butadiene/styrene monomer feed, whereas US6313232B1 focuses on tin-compound addition timing and does not disclose a specific styrene fraction in the extracted evidence.
- US12129321B2 and US9902837B2 both rely on water-based latex/emulsion processing and later coagulation/washing, but US12129321B2 is directed to porous battery-binder microspheres with very low styrene charge (25 parts styrene vs 75 parts butadiene in Example 1), while US9902837B2 targets oil-extended functionalized SBR for tire applications.
- US8614272B2 is a non-aqueous solution route with explicit monomer charge in moles and a post-polymerization end-modification step; it differs from the emulsion patents by using cyclohexane and precipitation/reprecipitation rather than latex coagulation.
- EP0295519A1 is process-control only: it discloses continuous SBR line blocking at 5-10 °C and solid-content control around 24.8-25.2%, but no monomer ratio or polymer composition target is provided in the extracted evidence.

4. CONCLUSION

Across the patent set, the most actionable low-styrene SBR synthesis guidance comes from emulsion and solution polymerization disclosures that explicitly control styrene feed. US8530017B2 provides the clearest emulsion-process range (10-50 wt% styrene), EP2657262B1 provides a solution-process benchmark (70/30 butadiene/styrene feed), and US9902837B2 shows that functionalized aqueous SBR can be prepared with bound styrene around 21 wt%. For low-styrene SBR development, the evidence supports using controlled styrene feed reduction in either emulsion or solution polymerization, with reactive emulsifiers or functional monomers used to preserve silica compatibility and end-use performance. Downstream coagulation, washing, stripping, and oil extension are recurring process elements that help convert the latex or cement into processable rubber.

5. REFERENCES

- US8530017B2 | Method for preparing styrene-butadiene copolymer using reactive emulsifier and ... | Korea Kumho Petrochemical Co., Ltd. | US | 2013 | <https://patents.google.com/patent/US8530017B2/en>
- EP2657262B1 | Triglyceride containing solution polymerization prepared styrene/butadiene ... | The Goodyear Tire & Rubber Company | EP | 2014 | <https://patents.google.com/patent/EP2657262B1/en>
- US6313232B1 | Continuous addition of tin halides to solution SBR | Bridgestone Corporation | US | 2001 | <https://patents.google.com/patent/US6313232B1/en>
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